RV COLLEGE OF ENGINEERING ®
BENGALURU-560059
(Autonomous Institution Affiliated to VTU, Belagavi)

"PathMind: An Adaptive Learning Path Generator"

Report

Database Management Systems

(CD252IA)

Submitted By

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Under the Guidance of

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in partial fulfillment for the award of degree of

Bachelor of Engineering

in

INFORMATION SCIENCE AND ENGINEERING

2025-26

RV COLLEGE OF ENGINEERING®, BENGALURU - 560059
(Autonomous Institution Affiliated to VTU, Belagavi)

DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING

CERTIFICATE

Certified that the Mini Project work entitled '**PATHMIND: An Adaptive Learning Path Generator'** has been carried out as a part of Database Management Systems Laboratory(CD252IA) in partial fulfillment for the award of degree of **Bachelor of Engineering** in **Information Science and Engineering** of the Visvesvaraya Technological University, Belagavi during the year **2025-2026** by **Sania Joseph (1RV23IS106), Shankar Patel K U (1RV23IS109),** who are bonafide students of **RV College of Engineering®**, Bengaluru. It is certified that all the corrections/suggestions indicated for the internal assessment have been incorporated in the report deposited in the departmental library. The report has been approved as it satisfies the academic requirements in respect of work prescribed by the institution for the said degree.

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Department of ISE, RVCE,	Department of ISE, RVCE,
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Name of the Examiners	Signature with Date
1. ______________	____________________
2. ______________	____________________

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DECLARATION

We **Sania Joseph, Shankar Patel K U,** are students of Fifth Semester B.E Department of Information Science and Engineering, **RV College of Engineering ®,** bearing **USN: 1RV23IS106, 1RV23IS109,** hereby declare that the project titled "***PATHMIND: An Adaptive Learning Path Generator***" has been carried out as a part of **DATABASE MANAGEMENT SYSTEMS (CD252IA)** by us and submitted in partial fulfillment of the program requirements for the award of degree in Bachelor of Engineering in Information Science and Engineering of the **Visvevaraya Technological University, Belagavi** during the year **2025-2026.**

Further we declare that the content of the report has not been submitted previously by anybody for the award of any degree or diploma to any other University

Place: Bengaluru	**Name**	**Signature with Date :**
Date: 07/01/2025	**Sania Joseph**	
	Shankar Patel KU	

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CHAPTER 1: INTRODUCTION

Online education platforms have transformed learning by providing flexibility and accessibility. However, most platforms follow a rigid and linear curriculum that does not adapt to individual learners' pace, prior knowledge, or learning preferences. This one-size-fits-all approach often results in learner disengagement and inefficient learning.

PathMind: An Adaptive Learning Path Generator addresses this issue by implementing an Intelligent Tutoring System (ITS) that dynamically personalizes learning paths using Reinforcement Learning (RL) and Knowledge Graphs (KGs). The system models academic content as interconnected concepts and intelligently recommends the next topic based on the learner's progress.

1.1Terminology

- **Intelligent Tutoring System (ITS):** A computer-based system that provides personalized instruction and feedback.
- **Reinforcement Learning (RL)**: A machine learning approach where an agent learns optimal actions based on rewards.
- **Knowledge Graph:** A graph structure representing concepts as nodes and relationships as edges.
- **Neo4j:** A graph database used to store and query knowledge graphs.
- **DQN (Deep Q-Network):** A reinforcement learning algorithm that combines neural networks with Q-learning.

1.2 Purpose

The primary purpose of this project is to design and implement an adaptive learning system that generates **personalized learning paths** for students. The system aims to analyze student performance, learning history, and prerequisite dependencies to recommend the most suitable next topic. By doing so, it ensures that learners progress through the curriculum efficiently and logically, enhancing both understanding and retention.

1.3 Motivation

The motivation for this project arises from the increasing demand for personalized education in modern learning environments. Traditional learning management systems do not adapt to individual learner needs, often resulting in poor engagement and learning inefficiency. With advancements in artificial intelligence and database technologies, it is now possible to build systems that adapt in real time. This project is motivated by the goal of leveraging these technologies to create a smarter, learner-centric educational platform.

1.4 Problem Statement

To design and develop an intelligent tutoring system that dynamically recommends personalized learning paths by analyzing learner performance and prerequisite relationships among concepts, using reinforcement learning and graph-based database techniques.

1.5 Objective

The objectives of the proposed system are:

- To provide personalized learning paths tailored to individual learners
- To optimize the learning sequence and reduce unnecessary repetition
- To store and track learner progress securely using a database
- To improve learner engagement through adaptive recommendations
- To demonstrate the effective integration of DBMS and AI techniques

1.6 Scope and Relevance

The scope of the project includes academic institutions, online learning platforms, and self-learning environments. The system can be extended to multiple subjects and domains beyond programming. The project is highly relevant in today's educational landscape and supports inclusive and equitable education, aligning with UN Sustainable Development Goal 4 (Quality Education).

CHAPTER 2 REQUIREMENT SPECIFICATION

This chapter describes the functional and non-functional requirements necessary for the successful implementation of the system.

2.1 Specific Requirements

2.1.1 Functional Requirements

- The system shall allow students to register and log in securely
- The system shall store student profiles and learning history
- The system shall represent curriculum topics using a Knowledge Graph
- The system shall recommend adaptive learning paths dynamically
- The system shall track mastered and pending concepts
- The system shall retrieve data efficiently using database queries

2.1.2 Non-Functional Requirements

- **Scalability:** The system should support multiple concurrent users
- **Reliability:** The system should function without failure during usage
- **Security:** Sensitive user data must be protected
- **Performance:** Queries and recommendations should be generated efficiently
- **Usability:** The interface should be simple and user-friendly

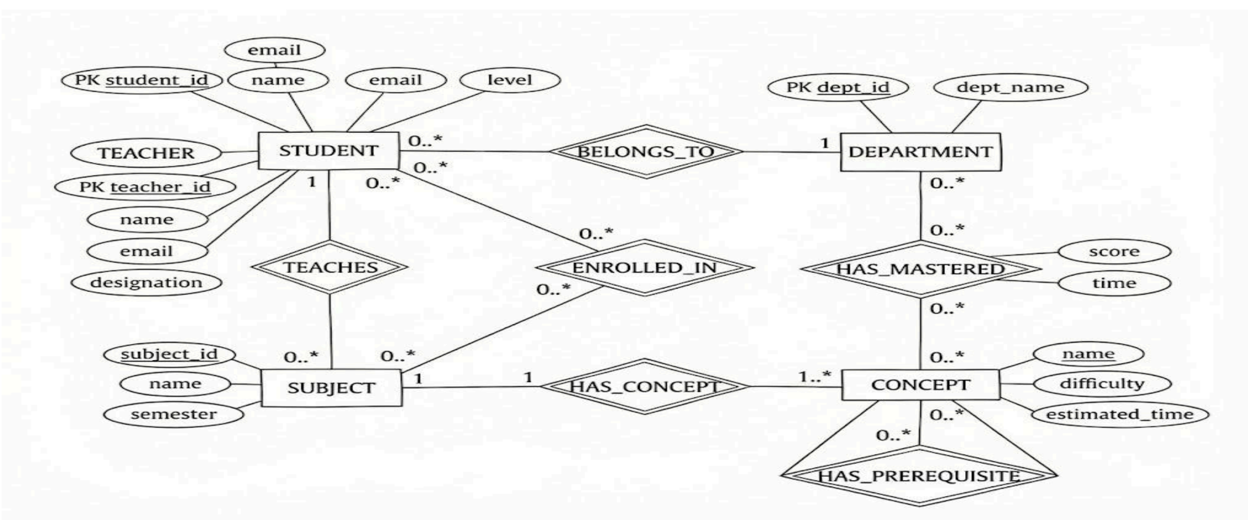
2.2 **Hardware Requirements**

- Minimum 8 GB RAM to support database and ML processes
- Intel i5 or equivalent multi-core processor
- GPU is optional but recommended for faster RL training

2.3 **Software Requirements**
- Python 3.9 or higher
- Neo4j Graph Database
- Flask Web Framework
- PyTorch or TensorFlow
- Pandas and NumPy libraries

CHAPTER 3 DESIGN

3.1 **E-R Diagram**



3.1.1 **Schema Representation**

The schema representation for the proposed system is derived directly from the Entity–Relationship (E-R) diagram by mapping each entity into a corresponding database structure. Since the system uses Neo4j, a graph database, traditional relational tables are represented as node labels, and associations between entities are modeled using relationships instead of foreign keys.

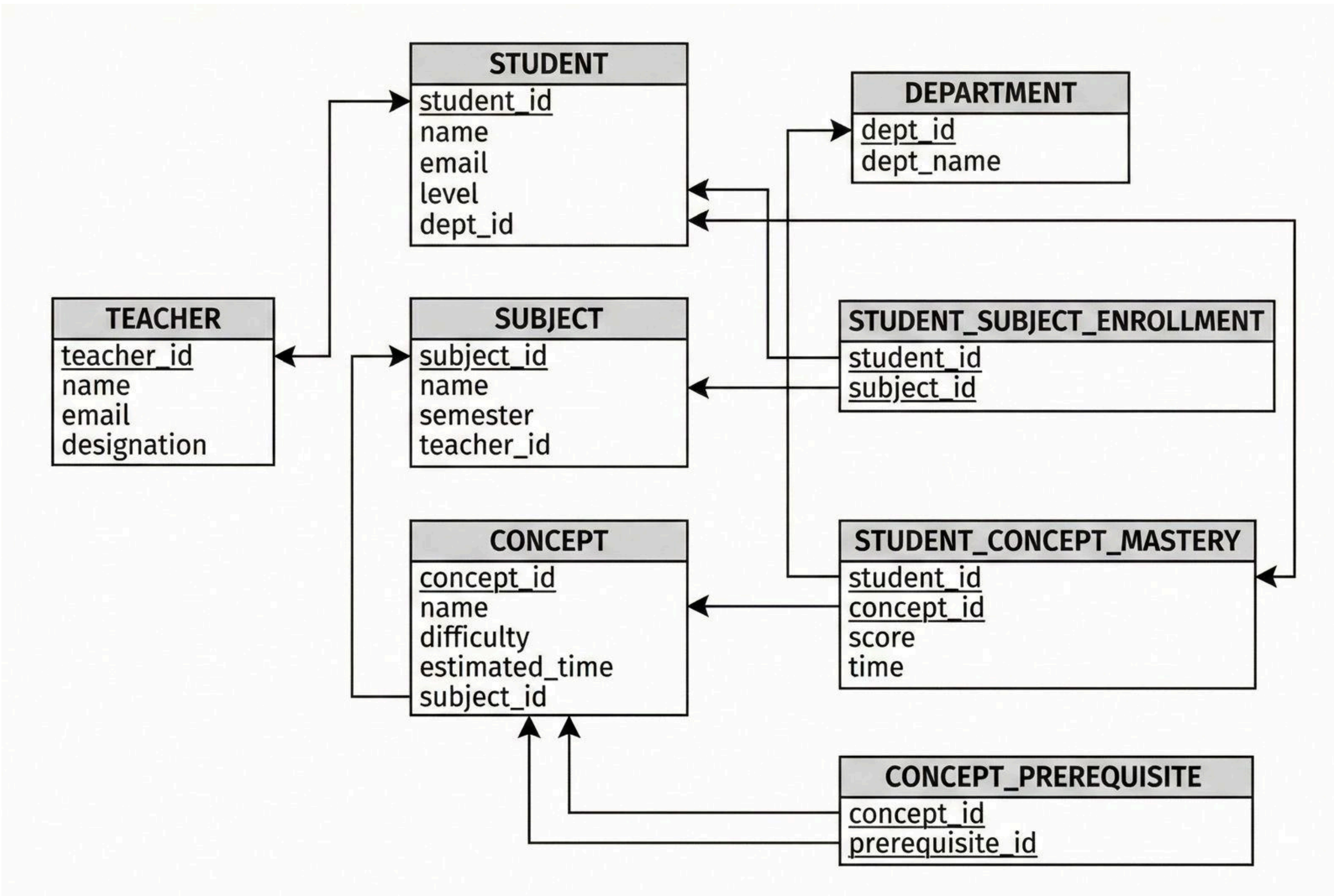
Each node is designed with a unique identifier to distinguish records, and relationships are used to represent dependencies and interactions between entities. This approach enables efficient traversal of prerequisite paths and learner progress tracking.

The Student node stores learner information such as student ID, name, email, password hash, and learning level.
The Teacher node stores instructor-related information including teacher ID, name, email, and designation.
The Subject node represents academic courses and contains subject ID, subject name, semester, and creation time.
The Concept node represents individual learning topics and includes attributes such as concept name, difficulty level, and estimated learning time.
The Department node stores department-related information and is linked to students for organizational grouping.

The relationships among these nodes capture the academic structure and learning workflow. For example, a student is enrolled in a subject, a subject contains multiple concepts, and concepts are connected through prerequisite relationships. This schema representation ensures structured data organization while enabling flexible and efficient querying in a graph-based environment.

Schema Representation (Node Labels):

- Student
  (student_id, name, email, password_hash, level, created_at)
- Teacher
  (teacher_id, name, email, password_hash, designation)
- Subject
  (subject_id, subject_name, semester, created_at)
- Concept
  (name, difficulty, estimated_time)
- Department
  (dept_id, dept_name)



3.2 **Normalization**

Normalization is an important database design process used to minimize redundancy and maintain data consistency. Although Neo4j is a non-relational database, normalization principles are still applied conceptually during the schema design to ensure clear separation of concerns and efficient data management.

Initially, data attributes that could cause duplication—such as subject details, department information, and concept metadata—are separated into distinct node labels. This prevents repeated storage of the same information across multiple entities. Relationships are used to connect related data instead of embedding repeated attributes within nodes.

Partial dependencies are avoided by ensuring that each node stores only attributes relevant to that specific entity. Transitive dependencies are also eliminated by separating logically distinct entities, such as keeping department information separate from student details.

As a result, the schema design effectively follows principles equivalent to **Third Normal Form (3NF)**, ensuring data integrity, scalability, and ease of maintenance.

3.2.1 **Schema after Normalization**

After applying normalization principles, the final schema consists of well-defined and independent node labels, each responsible for storing information related to a single entity. All attributes within a node are fully dependent on the unique identifier of that node.

For example, student-related information is stored exclusively in the **Student** node, while academic structure details are stored in **Subject** and **Concept** nodes. Prerequisite dependencies between concepts are represented using relationships rather than redundant data fields. Department information is maintained separately and linked to students through relationships.

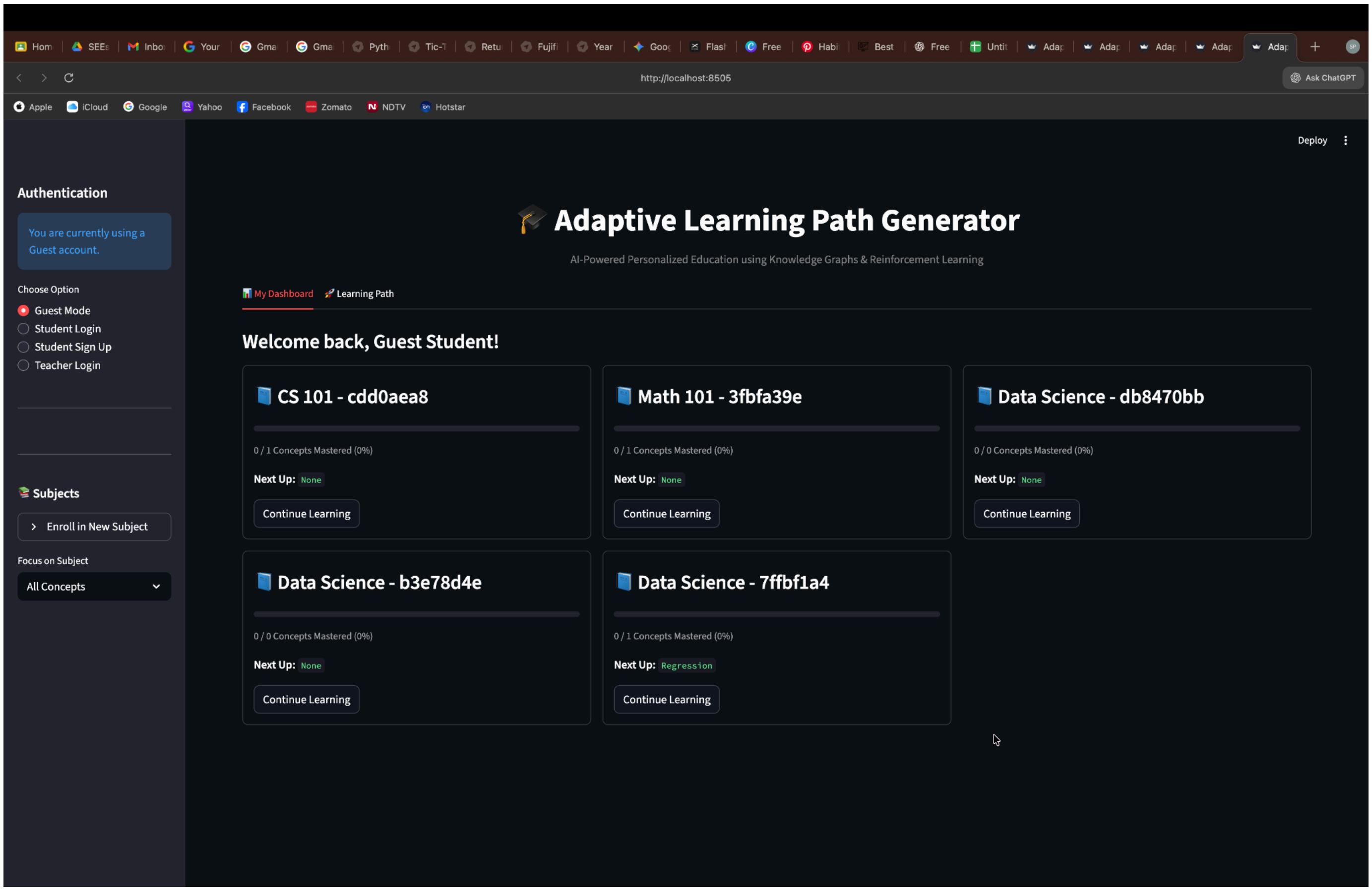
This normalized schema reduces data duplication, improves accuracy, and supports efficient graph traversal for adaptive learning path generation.

3.3 **Front End Design**

The front-end design of the system focuses on usability, simplicity, and effective interaction with the adaptive learning system. A web-based interface is developed to allow students and teachers to interact with the platform easily.

The interface includes forms for user registration and login, ensuring secure access to the system. Students are provided with dashboards that display their learning progress, mastered concepts, and recommended next topics. Teachers have access to analytical views that show student performance and concept-wise mastery levels.

All forms are designed with proper labels and input validation to reduce user errors. Visual elements such as progress bars, concept maps, and charts are used to present learning information clearly. The overall front-end design complements the backend database structure and enables smooth interaction between users and the intelligent tutoring system.



CHAPTER 4 IMPLEMENTATION DETAILS

4.1 Database implementation

4.1.1 Table Creation

Unlike traditional relational databases that store data in tables, the proposed system uses a Graph Database (Neo4j). In Neo4j, data is represented using nodes to model entities and relationships to represent associations between entities. This structure is highly suitable for modeling prerequisite dependencies and learning progress.

Nodes Created

1. Student: Stores learner-related information.
   Properties:
   student_id (UUID), name, email (Unique), password_hash,
   level (Beginner / Intermediate / Advanced), created_at
2. Teacher: Stores instructor-related information.
   Properties:
3. teacher_id (UUID), name, email (Unique),
   password_hash, designation

4. Subject: Represents an academic course.
   Properties:
   subject_id (UUID), name, semester, created_at
5. Concept: Represents an individual learning topic.
   Properties:
   name (Unique), difficulty (Easy / Medium / Hard), estimated_time
6. Department: Represents an academic department.
   Properties:
   dept_id, dept_name

Relationships Created

- (:Student)-[:ENROLLED_IN]->(:Subject)
- (:Teacher)-[:TEACHES]->(:Subject)
- (:Subject)-[:HAS_CONCEPT]->(:Concept)
- (:Concept)-[:HAS_PREREQUISITE]->(:Concept)
- (:Student)-[:HAS_MASTERED {score, date}]->(:Concept)
- (:Student)-[:BELONGS_TO]->(:Department)

These relationships allow efficient traversal of learning paths and progress tracking.

4.1.2 Table Population The database is populated using two different approaches:

1. Static Seeding
   The core curriculum structure, including all concepts and prerequisite relationships, is preloaded using a script named populate_full_curriculum.py. This initializes the Knowledge Graph structure (for example: Variables → Loops → Functions).

2. Dynamic Insertion
   - Users: Student and Teacher nodes are created dynamically when users sign up through the application.
   - Subjects: Subjects are created by teachers through the dashboard interface.
   - Mastery Relationships: A HAS_MASTERED relationship is automatically created when a student completes a quiz with a score greater than or equal to 70%.

4.1.3 Query Execution and Output

The backend interacts with Neo4j using the Cypher query language.

Example 1: Fetching Next Recommended Concepts

This query retrieves concepts that the student has not mastered yet and ensures that all prerequisite concepts have already been mastered.

```
MATCH (c:Concept)
WHERE NOT c.name IN $mastered_list
AND ALL(p IN [(c)-[:HAS_PREREQUISITE]->(pre) | pre.name]
    WHERE p IN $mastered_list)
RETURN c;
```

Output:
A list of eligible concepts such as ['Loops', 'Functions'].

Example 2: Teacher Analytics (Class Performance)

This query calculates the average score for each concept within a given subject.

```
MATCH (s:Subject {name: $subject_name})-[:HAS_CONCEPT]->(c:Concept)
OPTIONAL MATCH (c)<-[r:HAS_MASTERED]-(st:Student)-[:ENROLLED_IN]->(s)
RETURN c.name AS concept, avg(r.score) AS avg_score;
```

Output:
A JSON object mapping concepts to average scores, for example:
{'Variables': 0.85, 'Recursion': 0.60}

4.1.4 Security features

1. **Parameter Substitution**
   All Cypher queries use `$parameters` instead of string concatenation, preventing **Cypher Injection attacks**.
2. **Password Hashing**
   User passwords are never stored in plain text. The **bcrypt** algorithm is used to hash passwords before storing them in the database.
3. **Unique Constraints**
   Unique constraints are enforced on email properties to prevent duplicate user accounts.

4.2 Front End implementation

4.2.1 Form Creation

The frontend is developed using **Streamlit**, which enables rapid creation of interactive user interfaces.

- **Login / Signup Form:**
  Accepts Name, Email, Password, Department, and Level with validation checks.

- **Create Subject Form:**
  Allows teachers to create subjects and associate concepts using a multi-select dropdown.
- **Quiz Interface:**
  Dynamically generates a quiz with five multiple-choice questions and a submit button to evaluate performance.

4.2.2 Connectivity to the Database

The application follows a Layered Architecture:

1. Frontend Layer (app.py) – Handles UI interactions
2. Manager Layer (auth_manager.py, subject_manager.py) – Contains business logic
3. Database Layer (db_connection.py) – Manages Neo4j connections using the official Python driver

Data Flow:

User → UI → Manager Layer → Database Layer → UI

4.2.3 Report generation

Student Dashboard:

Displays progress using linear progress bars and a Graphviz-based concept map where:

- Green nodes represent mastered concepts
- Blue nodes represent recommended concepts

Teacher Analytics Dashboard:

Generates bar charts using st.bar_chart and highlights weak concepts (score < 75%) using alerts.

4.2.4 Security features

1. **Session Management:**
   Uses st.session_state to maintain login sessions.
2. **Secure Logout:**
   Clears session data using st.session_state.clear().
3. **Role-Based Access Control (RBAC):**
   Restricts access to pages based on user roles (Student / Teacher).

CHAPTER 5 TESTING AND RESULTS
5.1 Database Testing
5.1.1Test cases

Test Case ID	Test Description	Expected Result	Actual Result	Status
DB-01	Unique Email Constraint	Duplicate email rejected	Error shown	PASS
DB-02	Prerequisite Traversal	Concepts returned only if prerequisites mastered	Correct output	PASS
DB-03	Mastery Relationship	HAS_MASTERED created with correct score	Score = 0.8	PASS
DB-04	Subject Isolation	Student linked only to enrolled subject	Correct	PASS

5.2 Front EndTesting
5.2.1Test cases

TEST	Test Description	Expected Result	Actual Result	Status
FE-01	Empty Login Fields	Error message	Displayed	PASS
FE-02	Quiz Scoring	Correct percentage	80%	PASS
FE-03	Role Redirection	Correct dashboard	Loaded	PASS

5.3 SystemTesting
5.3.1Test case
Objective: Validate the complete learning cycle.

1. Student registers as a Beginner

2. System recommends an entry-level concept
3. Student completes quiz with high score
4. Concept marked as mastered
5. Next concept recommended automatically
6. Teacher dashboard reflects improved class average

Result: System functions as expected. **(PASS)**

CHAPTER6 CONCLUSION

The Adaptive Learning Path Generator demonstrates the effective integration of **Knowledge Graphs, Reinforcement Learning, and Database Management Systems** to deliver personalized education. The system dynamically adapts learning paths, ensuring mastery of prerequisites and improved learning outcomes.

6.1 Limitations

1. Static question bank
2. Internet dependency for database connectivity
3. Only MCQ-based quizzes supported

6.2 Future Enhancements

- AI-powered chatbot tutor
- Dynamic question generation using Generative AI
- Multimedia content integration
- Mobile application development

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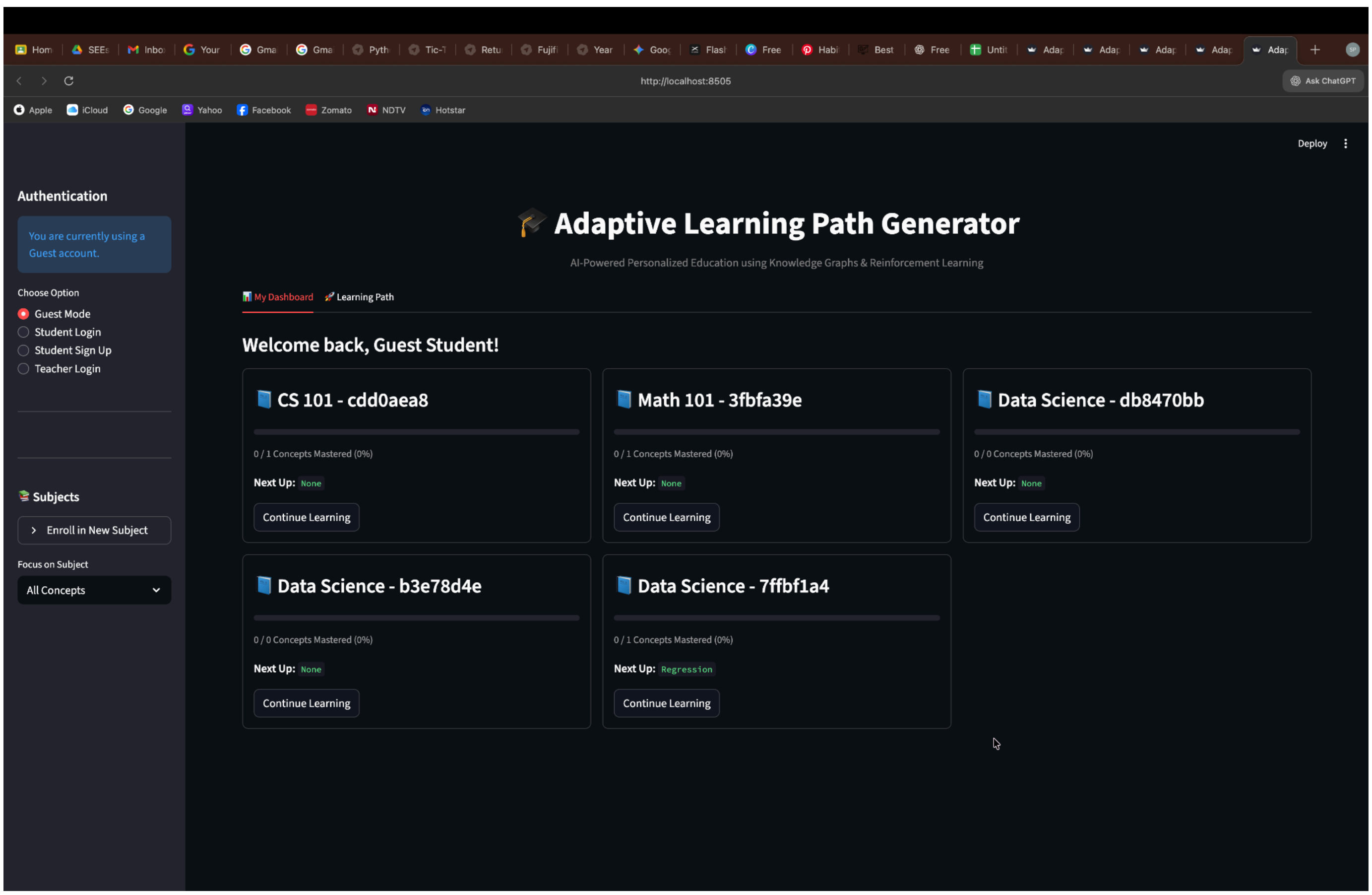
APPENDIX A- CODE SNIPPETS

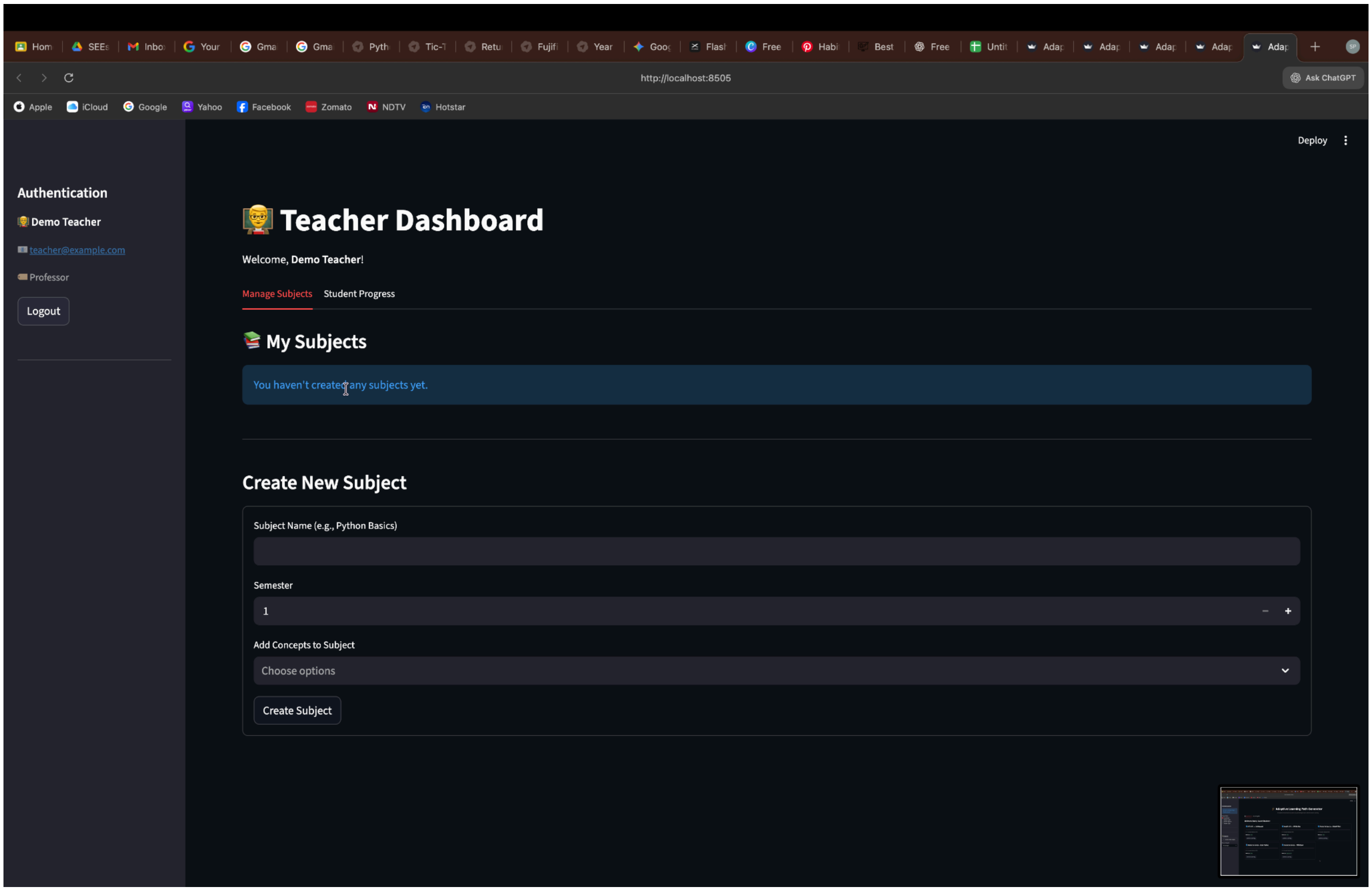
```css
/* CSS Custom Properties - Design Tokens */
:root {
  /* Colors - Light Theme */
  --bg-light: #FFFFFF;
  --bg-dark: #0D0D0D;
  --text-primary: #111111;
  --text-secondary: #666666;
  --accent: #00A8E8;
  --accent-hover: #008CCF;

  /* Typography */
  --font-primary: 'Inter', sans-serif;
  --font-size-h1: 56px;
  --font-size-h2: 40px;
  --font-size-body: 18px;
  --font-weight-regular: 400;
  --font-weight-semibold: 600;
  --font-weight-bold: 700;

  /* Layout */
  --max-width: 1200px;
  --section-padding: 72px;
  --border-radius: 6px;

  /* Spacing */
  --spacing-xs: 8px;
  --spacing-sm: 16px;
  --spacing-md: 24px;
  --spacing-lg: 32px;
  --spacing-xl: 48px;

  /* Transitions */
  --transition-fast: 0.2s ease;
  --transition-normal: 0.3s ease;
  --transition-slow: 0.5s ease;

  /* Shadows */
  --shadow-sm: 0 2px 4px rgba(0, 0, 0, 0.1);
  --shadow-md: 0 4px 8px rgba(0, 0, 0, 0.12);
  --shadow-lg: 0 8px 16px rgba(0, 0, 0, 0.15);
}
```

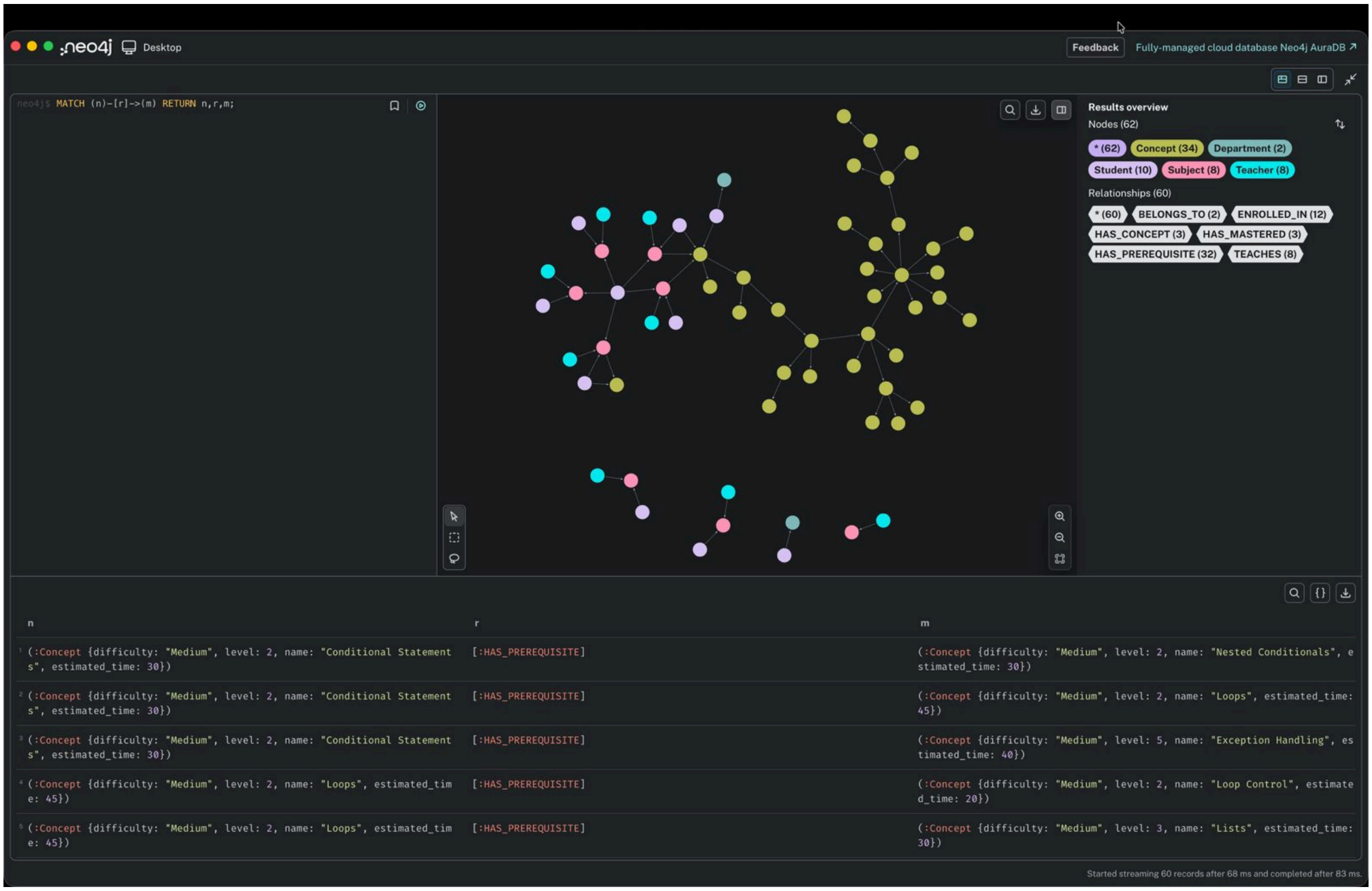
```
import { Suspense, useState, useRef } from 'react';
import { Canvas } from '@react-three/fiber';
import { OrbitControls, PerspectiveCamera, Environment, ContactShadows } from '@react-three/drei';
import './Model3D.css';

// Simple 3D Sachet Model using Three.js primitives
const SachetModel = () => {
    const meshRef = useRef();

    return (
        <group ref={meshRef} position={[0, 0, 0]}>
            {/* Main sachet body */}
            <mesh position={[0, 0, 0]} castShadow>
                <boxGeometry args={[1.5, 2.5, 0.3]} />
                <meshStandardMaterial
                    color="#00A8E8"
                    roughness={0.3}
                    metalness={0.1}
                />
            </mesh>

            {/* Top seal */}
            <mesh position={[0, 1.3, 0]} castShadow>
                <boxGeometry args={[1.5, 0.1, 0.3]} />
                <meshStandardMaterial
                    color="#008CCF"
                    roughness={0.4}
                />
            </mesh>

            {/* Bottom seal */}
            <mesh position={[0, -1.3, 0]} castShadow>
                <boxGeometry args={[1.5, 0.1, 0.3]} />
                <meshStandardMaterial
                    color="#008CCF"
                    roughness={0.4}
                />
            </mesh>

            {/* Label area */}
            <mesh position={[0, 0, 0.16]}>
                <planeGeometry args={[1.2, 1.8]} />
                <meshStandardMaterial
                    color="#FFFFFF"
                    roughness={0.2}
```

```html
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <link rel="icon" type="image/png" href="/logo.png" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <meta name="description" content="SnapFlow Beverages – Single-serve liquid beverages engineered for speed, convenience
    <meta name="keywords" content="beverages, hydration, single-serve, liquid drinks, SnapFlow, Insta-Barista, Hydro-Boost
    <meta name="author" content="SnapFlow Beverages" />
    <meta property="og:title" content="SnapFlow Beverages – Hydration Evolved" />
    <meta property="og:description" content="Single-serve liquid beverages engineered for speed, convenience & lifestyle.'
    <meta property="og:type" content="website" />
    <title>SnapFlow Beverages – Hydration Evolved</title>
  </head>
  <body>
    <div id="root"></div>
    <script type="module" src="/src/main.jsx"></script>
  </body>
</html>
```

APPENDIX B –SCREENSHOTS




http://localhost:8505
Ask ChatGPT
Apple
iCloud
Google
Yahoo
Facebook
Zomato
NDTV
Hotstar
Deploy
Authentication
Demo Teacher
teacher@example.com
Professor
Logout
Teacher Dashboard
Welcome, Demo Teacher!
Manage Subjects
Student Progress
My Subjects
You haven't created any subjects yet.
Create New Subject
Subject Name (e.g., Python Basics)
Semester
1
Add Concepts to Subject
Choose options
Create Subject


neo4j
Desktop
Feedback
Fully-managed cloud database Neo4j AuraDB ↗
neo4j$ MATCH (n)-[r]->(m) RETURN n,r,m;
Results overview
Nodes (62)
* (62) Concept (34) Department (2)
Student (10) Subject (8) Teacher (8)
Relationships (60)
* (60) BELONGS_TO (2) ENROLLED_IN (12)
HAS_CONCEPT (3) HAS_MASTERED (3)
HAS_PREREQUISITE (32) TEACHES (8)
n
r
m
1 (:Concept {difficulty: "Medium", level: 2, name: "Conditional Statements", estimated_time: 30}) [:HAS_PREREQUISITE] (:Concept {difficulty: "Medium", level: 2, name: "Nested Conditionals", estimated_time: 30})
2 (:Concept {difficulty: "Medium", level: 2, name: "Conditional Statements", estimated_time: 30}) [:HAS_PREREQUISITE] (:Concept {difficulty: "Medium", level: 2, name: "Loops", estimated_time: 45})
3 (:Concept {difficulty: "Medium", level: 2, name: "Conditional Statements", estimated_time: 30}) [:HAS_PREREQUISITE] (:Concept {difficulty: "Medium", level: 5, name: "Exception Handling", estimated_time: 40})
4 (:Concept {difficulty: "Medium", level: 2, name: "Loops", estimated_time: 45}) [:HAS_PREREQUISITE] (:Concept {difficulty: "Medium", level: 2, name: "Loop Control", estimated_time: 20})
5 (:Concept {difficulty: "Medium", level: 2, name: "Loops", estimated_time: 45}) [:HAS_PREREQUISITE] (:Concept {difficulty: "Medium", level: 3, name: "Lists", estimated_time: 30})
Started streaming 60 records after 68 ms and completed after 83 ms.